AI 101 for Mechanical Engineers · Module 8

Read metrics by their question

Lesson 8.2

What you will learn

> Calculate accuracy, precision, recall and F1, connect each metric to its question, and interpret a fault-detection example.

Lesson overview

Module 8 · Lesson 2

Read metrics by their question

AI 101 for Mechanical Engineers

Fault is the positive class	TP = 8 detected faults	FP = 2 false alarms	FN = 4 missed faults	TN = 86 correct healthy	N = 100 total examples

Metric	Question it answers	Formula and worked result
Accuracy	What fraction of all examples were classified correctly?	$(TP + TN) / N = (8 + 86) / 100 =$ **94%**
Precision	What fraction of fault alerts were actual faults?	$TP / (TP + FP) = 8 / (8 + 2) =$ **80%**
Recall	What fraction of actual faults were detected?	$TP / (TP + FN) = 8 / (8 + 4) \approx$ **66.7%**
F1	How do precision and recall combine?	$2TP / (2TP + FP + FN) = 16 / (16 + 2 + 4) \approx$ **72.7%**

Keep the counts visible

- 10 alerts: 8 faults and 2 false alarms.
- 12 actual faults: 8 detected and 4 missed.
- 94% accuracy alone hides the 4 missed faults.

Handle undefined cases

- With no positive predictions, precision has a zero denominator.
- Report the counts and the handling rule, not a silent value.

Know the limits

- F1 is a harmonic mean, pulled toward the smaller of precision and recall. It is not a safety requirement.
- ROC-AUC summarizes ranking across thresholds; for rare faults also check precision–recall and performance at the chosen threshold.

Key takeaway — **All these ratios are dimensionless. High accuracy alone does not describe fault detection.**

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Each metric answers a different question

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{N}$$
$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

- Accuracy: fraction of all N examples correctly classified
- Precision: fraction of fault alerts that were actual faults
- Recall: fraction of actual faults detected

F1 combines precision and recall

$$F1 = 2TP / (2TP + FP + FN)$$

- Harmonic mean of precision and recall, when defined [42]
- Pulled toward the smaller of precision and recall
- Does not encode costs of missed faults and false alarms

Worked example with N = 100

Metric	Calculation	Result
Accuracy	(8 + 86) / 100	94%
Precision	8 / (8 + 2)	80%
Recall	8 / (8 + 4)	≈ 66.7%
F1	16 / (16 + 2 + 4)	≈ 72.7%

TP = 8, FP = 2, FN = 4, TN = 86. Counts and ratios are dimensionless.

94% accuracy, yet four faults were missed

- Ten alerts; eight were actual faults
- Twelve actual faults; eight detected, four missed
- Accuracy alone does not describe fault detection adequately
- F1 is a summary, not a universal safety objective

Report zero denominators explicitly

- No positive predictions: precision’s denominator is zero
- No alerts give no evidence of useful precision
- Record the counts and the reporting rule
- Do not present an undefined quantity as a result

ROC-AUC summarizes ranking across thresholds

- Summarizes ranking behaviour via a ROC curve [41]
- A dimensionless discrimination measure
- For rare faults, also inspect precision-recall and class counts [43]
- Performance at the selected threshold still matters

> No single metric replaces operating-domain evidence or the consequences of errors.

CHECK YOUR UNDERSTANDING

For TP = 8, FP = 2 and FN = 4, what are precision and recall?

Answer before moving on.

CHECK YOUR UNDERSTANDING

For TP = 8, FP = 2 and FN = 4, what are precision and recall?

> Precision is 80%; recall is approximately 66.7%. They use different denominators: predicted alerts and actual faults respectively.

Key takeaways

- Accuracy, precision and recall answer different questions
- Include counts so missed faults remain visible
- F1 is a combined score, not a safety requirement
- Declare zero-denominator handling; assess relevant operating metrics

References

[41] T. Fawcett, “An introduction to ROC analysis,” Pattern Recognit. Lett., vol. 27, no. 8, pp. 861–874, 2006. doi: 10.1016/j.patrec.2005.10.010

[42] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” Inf. Process. Manage., vol. 45, no. 4, pp. 427–437, 2009. doi: 10.1016/j.ipm.2009.03.002

[43] T. Saito and M. Rehmsmeier, “The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets,” PLoS ONE, vol. 10, no. 3, Art. no. e0118432, 2015. doi: 10.1371/journal.pone.0118432

Numbers match the course reference list. The written lesson is the authoritative text.